

The Multiple Student Conceptions of the Chemical Bond in a Quantum Chemical Context

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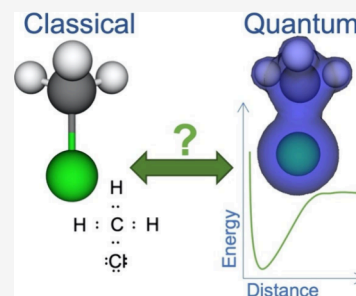
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ABSTRACT: The chemical bond is described with various context-sensitive models ranging from two electrons being shared between nuclei to the minimum-energy distance between the atoms. How we visualize these bonds further primes students to select one model over another. Previous research has targeted misconceptions of the chemical bond, but how they interact and evolve remains relatively underexplored. In this work, we examine the diversity of conceptions in bachelor's chemistry students who have started to learn about quantum chemistry. We present results of a thematic analysis of data produced by bachelor students at different stages in their studies while interacting with a chemical simulation triangulated with a metaphor analysis of interviews with students taking a quantum chemistry lecture. We found that the chemical bond was largely conceptualized either as a rigid entity or in relation to energy in writing and informally metaphorically as community. We argue that while these conceptualizations are valid models, the students struggle to understand the contextual dependency and the plurality of different models.

KEYWORDS: *Chemical Education Research, First-Year Undergraduate, Second-Year Undergraduate, Physical Chemistry, Misconceptions, Valence Bond Theory*



One aspect of initial quantum chemistry learning is the process of moving from classical to quantum conceptions. Many chemistry-specific concepts are introduced and highly simplified in a classical context, and these simplifications are only later explained or resolved. Student conceptions that lie at either end of this process have been well researched. For example, when discussing struggles surrounding chemical bonding, research focused on the different kinds of bonding, ionic, covalent, or metallic.^{1,2} From this perspective, the chemical bond is mostly conceptualized as a construct of sharing electrons between or across nuclei and is mostly represented with a line or two dots to indicate the localization of these quasi-electrons.³ In contrast, quantum chemistry education research focused strongly on the conceptions around atomic and molecular orbitals or valence bond theory and hybridization.^{4–6} Multiple misconceptions have been identified, mainly around the connection between the physical electron and the mathematical wave function, the derivation of molecular from atomic orbitals and the role of hybridization.^{3,7} However, research has focused on chemical-bond concepts within the context in which they are first learned (for example types of bonding in the context of an introductory chemistry lecture) as opposed to how these conceptions are activated in new contexts, for example in quantum chemistry learning.²

Here, we will not focus on the problematic aspects of teaching one model or another. We will also not discuss specific misconceptions that originate from these models. Rather, we are interested in what different models are (simultaneously) activated in the specific context of initial theoretical chemistry learning.

To this end, we triangulate data from four studies with first to sixth semester students. One of these studies was a usability study for an interactive quantum chemistry software, and two other studies primarily focused on investigating the effect of haptic feedback on learning about energy gradients. The haptic feedback corresponded to a change in energy, and the chemical bond was displayed based on distance criteria. We therefore assume that this novel technology did not impact the students' chemical-bond conceptions significantly. However, since we found interesting utterances by the students in an exploratory analysis of each of these studies, we decided to conduct a focused analysis on the chemical-bond conceptions as well as an additional triangulation with interviews. In analyzing their utterances and written statements on the chemical bond, we describe the different conceptions that were articulated in the specific context of software-assisted initial quantum chemistry learning.

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■ CONCEPTUAL FRAMEWORK

Conceptions

We apply the definition by Gilbert and Watts,⁸ who state that “conceptions’ be used to focus on the personalized theorizing and hypothesizing of individuals” (p 69). They can be accessed through actions of people, including speech and writing, but also nonverbal actions such as gestures. Naturally, not all conceptions are correct or accurate in the sense that they agree with current scientific thought (in the case of scientific conceptions). These incomplete or inaccurate conceptions have been termed alternative conceptions,⁹ or misconceptions.¹⁰ Often, they are accurate in the specific context in which they were integrated but are then wrongfully generalized to other contexts. How learners’ conceptions evolve toward expert knowledge, that is how conceptual change can occur, is theorized in two major lines of research—theory change¹¹ and knowledge-in-pieces.¹² The former is not to be confused with the similar term theory of change used in the field of evaluation to describe reforms in education.¹³

Both lines assume that small knowledge structures are crucial to conceptual change. Important for the presented work is the cross-domain mapping, which plays a role in both lines of reasoning. This process is characterized by the assumption that new information must be integrated into prior knowledge. The new information or target domain includes the target concept that is taught. What is included in the source domain depends on the theoretical framework that is applied. However, all of the mentioned accounts involve minimal knowledge structures that are often experiential and embodied. In both frameworks, misconceptions arise when (i) they are mapped inaccurately to the source domain or (ii) they are not actually accessible to the learner.¹⁴

For example, Venville and Treagust¹⁵ designed a study that introduced DNA metaphorically as a code, a metaphor that is commonly applied in textbooks, such as, for example, “cells replicate their DNA [... and] decode the instructions represented in the DNA” (ref 16, p 25). They found that whether this metaphor is productive for learning depends strongly on the learner’s conception of code. For example, one of their students potentially conceptualized code as a series of numbers and consequently did not conceptualize DNA as a code but rather as having a code, which resulted in different, potentially incorrect reasoning.

For chemistry specifically, Talanquer¹⁷ has suggested a model to classify and understand students’ misconceptions based on what he calls commonsense chemistry. We argue that his framework aligns well with the conceptual change framework described in this section. Students typically start their academic journey with more experience in everyday activities and common sense reasoning, and less experience in the chemical domain, especially due to the emergent nature of chemical knowledge structure, which can result in misconceptions.¹⁸ Therefore, these commonsense conceptions should be considered when introducing new concepts. Students are not presented with new information in a vacuum, but they will ground this information in prior experience or knowledge.

Here, we are interested in what chemical bond conceptions the students apply when they are first confronted with the potential energy surface and, thereby, with new tools to argue about stability and reactivity.

For this, we deployed a qualitative methodology that allowed us to give detailed descriptions and interpretive accounts of the

students’ experiences with the learning environment. To this end, a thematic analysis was employed. This approach offered the flexibility to create a priori codes based on the existing extensive body of research on misconceptions of the chemical bond as well as to create inductive codes during the analysis process.^{19,20} The results were triangulated with a metaphor analysis,²¹ as a large part of the domain-specific language in chemistry is of metaphorical nature. In the following, we elaborate on the basis for the a priori codes and thus the empirical literature on chemical bond conceptions of university students.

Chemical Bond Conceptions

The misconceptions that students hold of individual bonding models have been researched well with a strong focus on either the different kinds of chemical bonds² or hybridization.^{4,5}

As is often the case with incorrect conceptions, many misunderstandings can be traced back to either oversimplification or overgeneralization. For example, students often oversimplify ionic compounds by assuming that ionic bonds only exist between alkali metals and halogen atoms²² or only between single atoms.²³ Furthermore, current teaching methods tend to create the illusion that chemical principles and rules are exact and deterministic rather than a heuristic.^{24–26} For example, this overgeneralization of the octet rule can lead to misconceptions concerning chemical bonding.^{24–26}

Specifically, Boo found that grade 12 students often conceptualized the chemical bond as a physical entity.²⁷ Hence, students assumed that as constructing a physical object requires energy, so does the formation of a chemical bond. Similar evidence was also found in university students.^{22,28} The reason for this could be a manifold. For one, students struggle to grasp the concept of potential energy in the chemical domain, due to its prior introduction in multiple other contexts such as biology, everyday life or physics lectures.²⁹ Second, Niebert and Gropengiesser³⁰ argued that imperceivable concepts such as potential energy or the chemical bond must be understood through conceptual metaphor. This means that by visualizing chemical bonds as sticks, educators might inadvertently create the metaphor “Chemical Bond is Stick” and map the characteristic of a stick being physical to the chemical domain as well.

When confronted with the quantum chemical understanding of the chemical bond as a less rigid and discrete phenomenon, the students’ conceptions might be challenged without the necessary support to integrate the new information accurately. The traditional curriculum leaps from the chemical bond as a physical construct to one-electron wave functions and electron density. The quantum chemistry concept inventory developed by Pérez García and co-workers measured the understanding of the chemical bond concept in four items with one specifically focusing on the potential energy.³¹ In the administration of the test prior to the quantum chemistry lecture, which would correspond quite well to the prior knowledge level of the participants of our studies, many students showed the misconception that “[e]nergy is required to form the bond and that energy is stored within the bond”. This answer option remained popular, even after the quantum chemistry lecture. However, decreasing energy was successfully attributed to higher stability by most students on two items. This could indicate that the students did not yet associate chemical bonds with stability, an association that potentially

Table 1. Overview of Income Characteristics of Participants As Well As the Obtained Data for the Four Studies

Characteristic	Study I		Study II	Study III	Study IV
Semester	1st and 3rd	2nd		4th	6th
N_{total}	7 and 3	24		68	6
N_{used}	4 and 1	11		15	6
English fluency (conversational, fluent, native)	0, 3, 2	1, 8, 2		2, 10, 3	—
Gender (f, m, x, —)	1, 4, 0, 0	4, 7, 0, 0		7, 7, 1, 0	—
Age	20.6 $\pm$ 3.0	19.7 $\pm$ 1.4		20.9 $\pm$ 1.2	—
Data	Think-aloud	Written answers to open-ended questions		Written answers to open-ended questions	Interviews

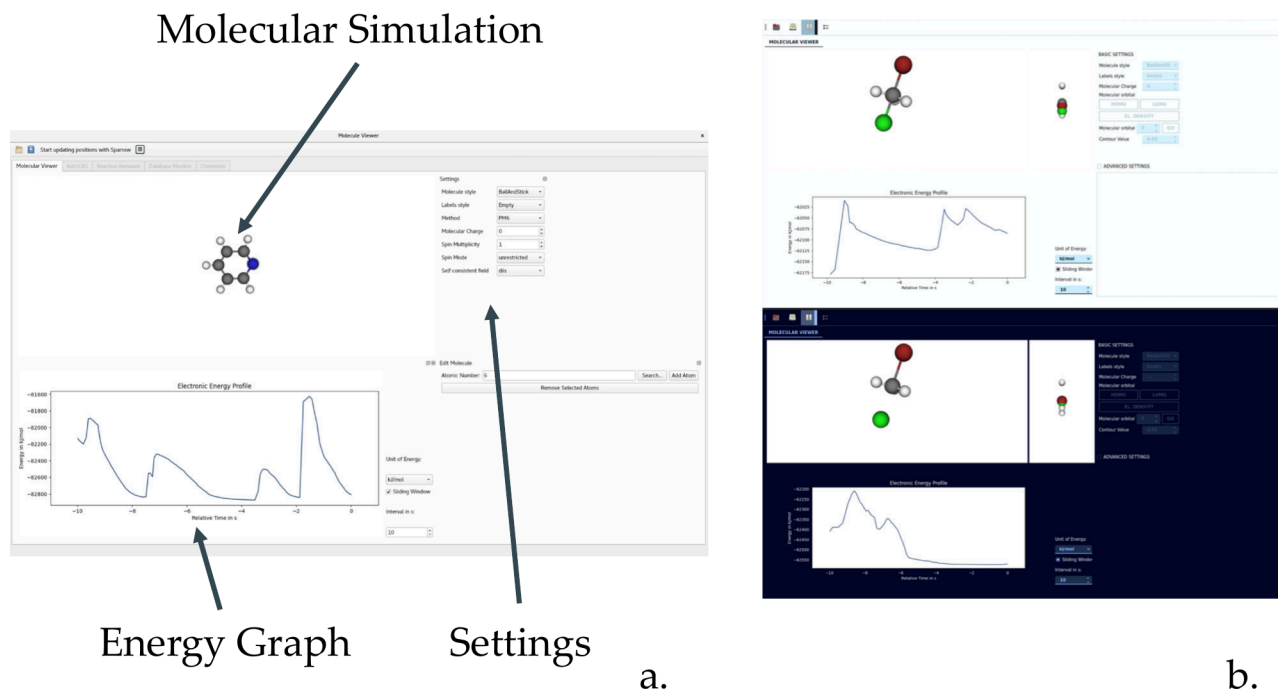


Figure 1. SCINE Heron^{39,40} at the stage of the usability user study (a) and at the preparation for future learning studies (b). In both versions, the chemical bonds are displayed as sticks based on distances between atoms.

could be attributed to the physical entity conception of the bond.

Moreover, studies show multiple misconceptions around atomic and molecular orbitals and hybridization.^{4,5} As a potential facilitation for this transition, it was suggested to focus on the potential energy diagram of bonding formation.^{32,33} In particular, this approach highlights the discrepancy between a physical entity that requires energy to be built and the chemical bond that lowers the Born–Oppenheimer potential energy of the molecular system when being formed.

In summary, the literature suggests that the students show misconceptions both for classical depictions of the chemical bond and for concepts connected to the chemical bond in quantum chemistry such as orbitals or electron density. Therefore, the construction of quantum conceptions might be hindered by or blended with existing classical conceptions. We investigate this process further by analyzing and comparing utterances and written statements of university students at the beginning of the first semester until students who visit the first quantum chemistry lecture in the sixth semester.

In Switzerland, the chemical bond is traditionally introduced to the students at the high school level. The students first learn about the covalent bond, potentially connected to Lewis structures, followed by ionic and metallic bonds. Conse-

quently, they first encounter the chemical bond as a structural entity. In the first year of the bachelor program, the chemical bond is explicitly described in the organic chemistry lecture with the Lewis model and refined with the concepts of polarity and resonance theory. The chemical bond is less prominently mentioned in inorganic and physical chemistry lectures. However, the physical chemistry lecturer relates the chemical bond to the molecular orbital model. The chemical bond concept is then applied and refined in many lectures, the full review of which is out of the scope of this manuscript. Potentially important for the given context, chemical bonds are not needed in computational chemistry calculations. The latter solely rely on the coordinates of the nuclei.

In this work, we pose the following research question: What conceptions are articulated by bachelor students?

METHODS

We combined qualitative data from three studies with university students and triangulated the data with an interview study. First, think-aloud data from a usability study (study I, first and third semester students), second, written answers to open questions from two preparation for future learning (PFL,^{34–37} study II (second semester students) and III³⁸ (fourth semester students)) studies and third, an interview

study (study IV, sixth semester students) were analyzed. In the following, the studies are described and the analysis process is specified.

Participants

All studies were conducted with bachelor students from ETH Zurich who were enrolled in a program strongly related to chemistry (biochemistry, chemistry, chemical engineering, or interdisciplinary sciences). An overview of the incoming traits and corresponding data is given in Table 1. The students' chemical proficiency ranged from being a first semester student in a general chemistry lecture without any prior knowledge in quantum chemistry to sixth semester students attending their first quantum chemistry course. The participants for studies I and II were recruited on a voluntary basis via E-mail. For studies II and IV, we (additionally) advertised the study in a lecture. Participants of study III participated in the study as part of a course. For all studies, the data were only used if the student gave informed consent. All studies were approved by the ethics committee of the university with proposal EK-2021-N-41.

Materials

All studies employed the same learning environment, SCINE Heron (Figure 1). SCINE Heron^{39,40} is an open-source graphical quantum chemistry environment that allows interaction with molecular systems while the latter responds in real-time. On the screen, the students saw a potential energy plot that showed the energy of the molecular system in real-time. The students could interact with the molecular structure by dragging atoms across the screen and hence explore different reaction mechanisms. The rest of the structure simultaneously relaxed according to an energy criterion in ultrafast quantum chemical calculations.^{41–43} Thus, the atoms continuously relaxed toward a minimum-energy structure. In addition, the students felt haptic feedback in the form of resistance or attraction that corresponded to the calculated quantum chemical force that acted on the picked atom through a Touch haptic device. Important for the presented work is that the chemical bonds were represented by sticks and the atoms by balls on the screen. A bond was displayed based on the distance between atoms. This choice of visualization meant that (i) the breaking and forming point of the bond was somewhat arbitrary from a chemical point of view, and (ii) one atom could form arbitrarily many bonds as opposed to as much as the octet rule would allow for molecules made from second-period atoms.

The participants of study I interacted with a pilot version of the final interface, while the participants of studies II to IV both interacted with the final product. While the visuals differed in some aspects as can be seen in Figure 1, the visualization of the bond did not change. In all studies, the tasks were very similar and will be described in the Procedure section. The goal was to compare different reaction mechanisms and connect them to energetic evolution. While the focus lied on the reactions, the chemical bond still played an important part as specific bonds were supposed to be broken and formed. In study I, the students were asked to think aloud while interacting with the environment. The instructor did intervene as little as possible and spoke only to remind the students to think-aloud.

In studies II and III, the students were presented with open-ended questions about the instructions. The questions were supposed to elicit contrasting and comparing as well as

reflection. Examples of these questions are “What differences or similarities did you notice?”, “Are your conclusions consistent with what you observe?”, and “What changed?”. The prompts referred to different reaction mechanisms that the students tried to carry out. They were answered in writing directly on the instruction sheets. Here, although not explicitly named, the focus lied on the Born–Oppenheimer potential energy, the force that they might have experienced, and the reaction mechanisms. Still, we found many notions of the chemical bond that we analyze later in this work.

The data that resulted from these studies was triangulated with the interview data of study IV. The students were asked first about a definition for the chemical bond and were then presented with three different visual representations of the same nucleophilic substitutions (Lewis structure, ball-and-stick model, and electron density model, see Figure 2), before they performed the reaction with SCINE Heron.

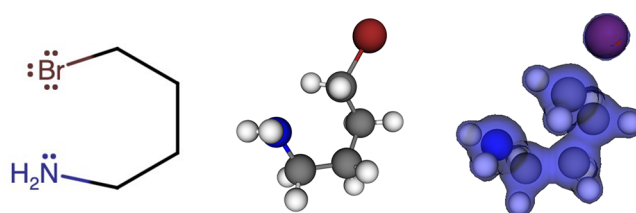


Figure 2. Products of the nucleophilic substitution reaction were presented to the students. The visual representations further included the reactants and the double arrow indicating the reaction, but no electron-pushing arrows. The students were asked to hypothesize a reaction mechanism and reason about driving factors.

Design

A summary of the study designs is given in Figure 3. The data analyzed stem in all three cases from a problem-solving phase. In the case of studies II and III, this was followed by an instruction phase, a design commonly known as problem-solving followed by instruction (PS-I^{35,44–46}). As for the presented analysis, the consequences of this design choice are

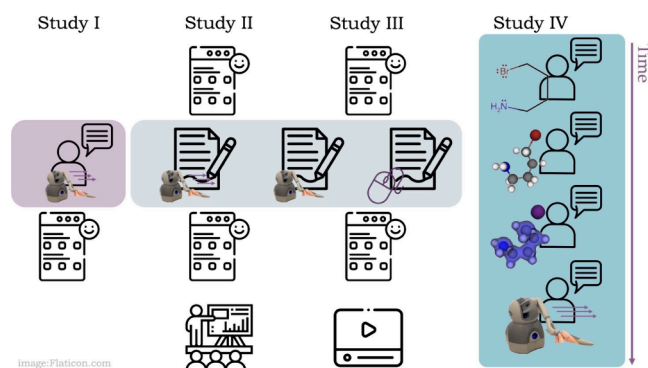


Figure 3. Overview of the designs of the studies providing the data analyzed in this work. The qualitative data collection is indicated with a colored background. For studies I–III, this included the problem-solving tasks. Students interacted with a software that continuously relaxed the molecular structure on the screen toward an energy minimum. Their task was to create a product from the reactants while feeling the force that would act on the atom they selected. For study IV, this included the entire interview data. This figure has been designed using images from Flaticon.com.

not further important, we refer the interested reader to the published study III.³⁸

While in study I, the focus was on identifying technical problems of the program and usability issues, in studies II and III, the focus was on learning according to the PS-I design principle. To implement this, we chose an exploratory approach. When breaking bonds in the first task, the students became aware of the connection between the potential energy and the molecular structure. Then, as the second and third tasks followed the same reaction mechanism, the students could apply this connection to contrast, generalize, and identify deep features of chemical reactivity—mechanism elucidation as minimal energy problem, the role of molecular and partial charge, and the role of frontier molecular orbitals. The data analyzed in this work was gathered during the problem-solving phase only, where the procedure was identical for studies I–III.

The interviews (study IV) were semistructured with the structure being provided by a sequence of differing visual representations (see Figure 2).

Procedure

The procedure of the problem-solving phase was almost identical for the first three studies. They differ only in the qualitative data obtained. For study I, the students thought aloud during the process, while in studies II and III, they answered open-ended questions in written form directly on the instruction sheets. The quantitative data obtained in each study differed based on the goals of each study, but as this does not concern our work here, we omit the details.

Consent was sought from all students before the start of the studies. After the first phase of quantitative measures, the students performed the problem-solving phase. The problem to be solved was the execution of a reaction, that is, the creation of a specific product from given reactants. The experience started with rotation, zooming, and dragging and dropping tasks to familiarize themselves with the Touch haptic device. After self-proclaimed proficiency, the students moved on to the learning tasks. These three tasks consisted of (i) a single molecule that they could manipulate as they pleased to explore the connection between energy change and force and (ii) two nucleophilic substitutions with differing reactants. In addition to exploring different reaction mechanisms, the students could also explore the effect of molecular charge or visualize orbitals and electron density. After the tasks were finished, the students again filled out questionnaires for the quantitative measures before receiving the instruction phase (for studies II and III) or being finished (for study I). All studies took between 90 and 120 min. During the whole process, the instructor aimed to interrupt as little as possible. Still, there were some technical issues, and the students sometimes struggled with the three-dimensional navigation of the haptic device, with which the instructor then assisted. The reader interested in the procedure in more detail may refer to Müller and co-workers (2024).³⁸

The interviews of study IV started with a short introduction of the participant mentioning their background, preferred and unfavored classes, semester projects, and plans for bachelor or master theses. Subsequently, they were asked to give a definition of the chemical bond. Afterward, the students were presented with three visual representations, successively (see Figure 2). For each representation, they were asked about the reaction mechanism, potential factors that are driving this mechanism, and the role of the chemical bond in the given

representation. In a final task, they could perform the reaction with a SCINE Heron. The interviews took between 50 and 70 min. The procedure is explained in more detail in the Supporting Information.

Analysis

As we were interested in the complexity of views that students bring to and have of the learning experience, the resulting qualitative data was analyzed using an experiential realist, constructionist perspective with a semantic orientation.^{20,47} Specifically, we assumed that the students' knowledge of the world emerges from active experiences in the world. "[T]ruth is not external to the human mind. Rather, truth is the product of the human mind" (ref 47, p 230). This is in contrast with the postpositivist view that there is one true learning trajectory or at least a set of probable learning processes to be uncovered.⁴⁸ The first author conducted the analysis alone.

We accounted for Lincoln and Guba's^{49,50} four dimensions of trustworthiness (credibility, dependability, transferability, and confirmability) as follows. We triangulated several data sources (think-aloud data, written answers to open-ended questions, and interviews) and analysis methods (thematic and systematic metaphor analysis) to increase credibility. Furthermore, the first author who conducted the analysis discussed her findings and reflective notes with the other authors. Some of her reflections with regards to her biases and shift in epistemology are reported in the Supporting Information to increase dependability. By relating our findings to other studies in chemistry education research, we are pointing toward avenues for possible transfer, and while we do not aim for objectivity, we hope that the number of examples provided in the Results section as well as in the Supporting Information (including raw data in German) will allow the interested reader to confirm or retrace our findings.

The analysis process can be divided into two phases. The initial thematic analysis²⁰ of studies I–III resulted in an exploratory account of chemical bond conceptions. These were analyzed in more detail and were later triangulated with a metaphor analysis²¹ of interview data targeting the chemical bond concept specifically.

In studies I–III, we started with certain a priori codes that we expected based on the specific conceptual framework of the study. We then coded the data in a fine-grained manner, developing codes as needed and allowing an unlimited number of codes per segment to avoid missing any inductive meanings. This process was continued until the full data set was coded, and no further codes were identified. The codes were then grouped into topic summaries, from which themes were derived.

In study I, we focused on the usability of the environment and whether the environment was interpreted as intended. The a priori codes of studies II and III were grounded in the theoretical framework of embodied learning⁵¹ and focused on how the haptic feedback was perceived and whether the visual and haptic representations were connected and interpreted by the students as intended. In addition, we were interested in inductive themes around the interpretation of the learning environment.

In all three studies, we found that the students brought a wide variety of conceptions concerning visual representation of the molecules. We found that the visual representation of the chemical bond as sticks that were displayed depending on the interatomic distances could confuse the students and even

Table 2. Themes and Specific Codes

Theme	Codes	Description
The chemical bond as rigid entity	• Ability to form bond • Bonding as spontaneous • Different kinds of bonds • Expectation of event at bond breaking or formation	Breaking and forming of a bond is a sudden event. The bond is either there or not, it is a physical entity that is created and destroyed. To do that, the two atoms need to have the ability to form this physical entity.
The chemical bond related to energy	• Activation energy • Bond as minimal energy distance • Bonding lowers energy • Other relation of energy to bonding	Breaking and forming of a bond is a process of moving toward a minimum-energy structure. Hence, the final stage of the bonding process reduces the potential energy of the molecular system. That being said, an activation barrier needs to be overcome to arrive at the product.
Orbitals as indicators of the presence or stability of the bond	• HOMO = instable bond • No orbital, no bond • Orbitals separation is bad	As orbitals describe electrons and electrons are the substance of a chemical bond, the orbitals describe the bond now. They can therefore also indicate stability of a molecule. Almost all conceptions found in this theme were inaccurate.

make them mistrust the simulation. As we found this in all studies although the analyses focused on different goals, we decided to explore further and thus combined the topic summaries around chemical bond conceptions and reanalyzed them. For this second analysis iteration, we were interested in what conceptions of the chemical bond might be activated or challenged in the learning environment and how they might have been influenced by the chosen visual representation. We expected known conceptions such as the different kinds of bonds, ionic, covalent and metallic, possibly an understanding in terms of energy that breaking and forming bonds might be related to energy increase and lowering, respectively.² As we were again following a thematic analysis approach, we were open to further conceptions. All segments coded with a chemical bond related code were recoded according to the model that they activated. These codes were then organized into themes that are presented in the Results section.

The interview data was analyzed using the systematic metaphor analysis approach as introduced by Schmitt.²¹ First, the relevant utterances mentioning the chemical bond were identified. Second, the utterances were analyzed for metaphors. Third, the metaphors were grouped into metaphor concepts. Though the analysis was performed with the results of the thematic analysis already known, we attempted to look for metaphors beyond the identified themes.

For reporting, the answers were translated from German to English, including possible mistakes made by the students. Original transcript excerpts are provided in the Supporting Information.

RESULTS

Exploratory Thematic Analysis

The analysis process is described, and further data examples are given in the Supporting Information. The final codes that were organized into themes are presented in Table 2. We generated three themes that capture how the students conceptualize the chemical bond: (i) the chemical bond as a rigid entity, (ii) the chemical bond related to energy, and (iii) orbitals as indicators of the presence or stability of the bond. First semester students in particular, but also students of other semesters, tended to focus on the capability of an atom to form

a bond and what kind of bond might be formed, hence “The chemical bond as rigid entity” (see Figure 4). For example, J.

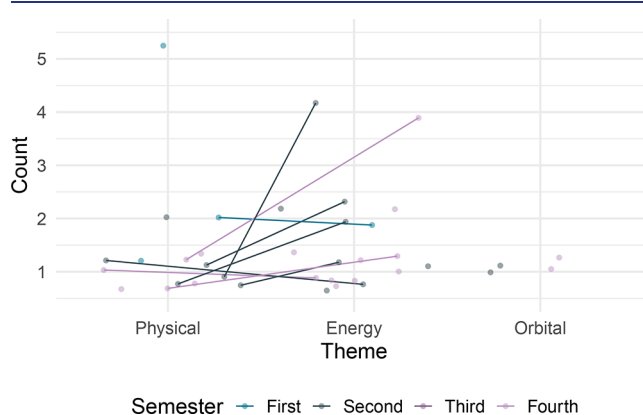


Figure 4. Count of codes assigned to the generated themes. This is to give an impression of the distribution of the different statements and written answers but should by no means be interpreted as statistical evidence. Individual students are represented by a point if they were situated in only one theme or as a line if they used more than one model throughout the intervention.

A. of the first semester asks whether “hydrogen actually [is] able to form this bond”. Some of these observations were activated by the visualization such as “Bond extends until it breaks” (C. X., second semester) or “I have the feeling that I can pull both atoms equally much away before the bond breaks” (E. E., second semester). Some students even showed confusion when nothing specific happened or changed upon the event of formation or breaking of a chemical bond.

“So, I am doing this and then, when it forms a bond here, absolutely no difference.” (N. Y., first semester)

“E rises the closer I move Cl to C even before + after bonding” (M. M., fourth semester; E refers to the energy graph)

We attribute this to a certain extent to the ball-and-stick representation of the molecular system. As breaking a stick in the macroscopic world means a sudden event, the students might have translated this characteristic to the chemical world. In addition, this might be supported by other binary languages

Table 3. Metaphor Concepts and Individual Metaphors Translated from German

Metaphor concept	Metaphors	Example
Bond is Community	• Bond is Sharing	• So for me, it [the chemical bond] is if two atoms somehow share electrons.
	• Electrons are Baggage	• [...] so, he [the nitrogen] has a smaller charge [in German the same word as Baggage "Ladung"] than the carbon over there.
	• Bond formation is Connection	• [...] this has many electrons that means, it will enter somehow a relationship or so. [in German, "eine Bindung eingehen" means both forming a bond and entering a relationship]
	• Electrons are People (in love)	• Electron pair
	• To be part of a molecule is Belonging	• And then it [an atom] leaves.
Bonding is Construction	• Forming a bond is Building a bridge	• It is the smallest energy gap, quasi, that one has to bridge.
	• Molecule is Object that can be destroyed	• Ah, maybe, the bromide is split off.
	• Bond is Structural component	• I do not know, why this bond to the bromide is formed.
	• Interaction is Glue	• That is [the interaction] is what keeps it together, [...]
	• Bond is Chain link	• [...] we have one carbon less, the chain is shorter and [...]
Bond is Overlap	• Bond is Overlap	• [A bond is] when two orbitals overlap and have electrons each [...]

commonly employed in chemistry, such as the "breaking" of a bond or the "formation" of a bond. To a chemist, this "breaking" implies a process, but in colloquial language, "breaking" is associated with a sharp, brief, and quick event.

The second theme "the chemical bond related to energy" captures conceptions such as connecting an increase or decline of energy to the molecular structure or the chemical bond as a minimal-energy distance. This theme allows chemical bonding to be a process instead of a sudden event, although students also articulated their views in both themes at the same time, for example, when they expected a sudden change in energy at the point of bond formation (see excerpt by N.Y. above, the expectation is made explicit by the word *even*). This blending of themes is in agreement with the work by Becker and Cooper⁵² who found that science students conceptualized potential energy as capability for a structure to change, stored energy, or as related to stability. Hence, if a student conceptualized potential energy as capability (for bond formation) and the chemical bond as a rigid entity in which energy is stored, the physical entity and energetical understanding of the chemical bond will not necessarily contradict each other. Examples of this theme include

"Energy increases with distance of Br⁻ & decreases when Cl⁻ bonds [binds]." (C. M., second semester)

"The bonding enthalpy is released when a bond is formed." (M. K., fourth semester)

"If I come closer it raises [increases] until it ends up bonding and stabilizing." (F. P., fourth semester)

"As a bond appears, the energy falls again until reaching a minimum at a specific distance." (A. T., fourth semester)

Finally, a few students conceptualized the "Chemical bond in connection to orbitals". We found this conceptualization only in four students, and even these utterances were potentially activated in a biased way as we explicitly asked them to explore actively the orbital visualization feature of the environment. Still, we find it worth mentioning as it illustrates the difficulties moving from an "electron as particle" view to an "electron as described by a wave function" view. These students understood that their conception of a chemical bond is potentially incomplete, as they could not rely on a localized electron anymore. Therefore, they tried to connect the new information—the electron is described by a wave function or

orbital—and the known information—a bond is where two electrons are between atoms.

"Yes, because ipso to the amino group exists no LUMO [lowest unoccupied molecular orbital], therefore there cannot be a bond formed there." (B. E., second semester)

In this example, B. E. expects that there needs to be some space in which the electron must be able to move. Orbitals as regions in space for electrons to go into as opposed to mathematical models describing the electron is a known misconception.⁵ As the concept "electron" changes so much during chemistry education, it is of high importance to consolidate related concepts such as the chemical bond accordingly.

In general, we found that some students employed only a single model. Others employed one model over another at different stages of the problem-solving. Finally, there were students who employed multiple models in a single utterance, indicating potentially an attempted integration of the models. This highlights the unawareness of the context-dependent nature of the models. Although not representative due to the limited amount of data, there is an indication that the students start with mainly the physical entity conception and then learn about the molecule as a minimum-energy structure. First, they struggle as they try to integrate this new information into the chemical bond as physical entity model—as indicated by the multiple second semester utterances that use both models at the same time.

"Both graphs show a peak in energy prior to the breaking of the bond. Chloride seems to leave more easily." (S. X., second semester)

"If a bond is formed, the change is insignificant. [...] The bump is because the Br–C bond is cleaved and the steep slope because the Cl–C bond is formed." (I. D., second semester)

S. X. indicates with an event (the peak) *prior* to the breaking of the bond that the breaking itself is an event as well. An indication of timing is meaningful only if bond breaking is an event as opposed to a process. However, they also mention *leaving*, a verb that indicates an action and therefore a process. Similarly, in the second quote, *bump* denotes a sudden event at which the bond is broken. However, the formation of the bond is described as a process connected to the slope in energy.

Finally, in the fourth semester, they start to learn when to apply which model.

Directed Metaphor Analysis

The backgrounds of the interviewed students were quite heterogeneous and hence representative of a typical university classroom. Although the course is officially offered to sixth semester students, the cohort consisted of one-third year undergraduate chemistry student who would like to specialize in applied theoretical chemistry, three master chemistry students who are interested in computational chemistry calculations to support experimental research, and two interdisciplinary science bachelor students, of whom one had a strong background and interest in quantum physics and the other had a background in biology but plans to focus more on computational chemistry in the future. We acknowledge the bias that these students were motivated and ambitious enough to participate in a voluntary study and might hence only be representative of the cohort in terms of background but not in terms of interest and motivation.

The metaphor concepts are given in Table 3. Every metaphor concept is constructed from multiple metaphors found in the interview data. This ensures a stronger argument than solely considering individual metaphors. Next to the metaphor concepts presented in Table 3, we have found many occurrences of personification of concepts ([Concept] is the Agent). While it would be highly interesting to explore how this personification affects reasoning, this is beyond the scope of the present work.

Two of the three themes presented earlier, “The chemical bond as rigid entity” and “Orbitals as indicators of the presence or stability of the bond”, were found here as well in the metaphor concepts “Bonding is Construction” and “Bond is Overlap”. However, in this analysis of colloquial language, we further found a frequent notion of the bond as a community that involves sharing, connecting, and belonging.

DISCUSSION

Thematic analysis of the data of three studies investigating the effect of haptic feedback on learning quantum chemical concepts resulted in the common theme that chemical bonding is conceptualized in various models. Here, we reanalyzed the data related to the chemical bond and generated three themes that describe how students conceptualized the chemical bond in this particular context. The results of this analysis are triangulated with a metaphor analysis of interviews, which resulted in similar findings. In the following section, the reasons for similarities and differences are discussed.

We found that chemical bonding was viewed as an event (“the chemical bond as rigid entity”) by 15/31 (3 first, 6 second, 6 fourth semester) students and as a process (“the chemical bond related to energy”) by 19/31 (1, 8, 10) students with 9 (1, 5, 3) students using both. The third theme “Orbitals as indicators of the presence or stability of the bond” was found substantially less frequently (mentioned by only four (0, 2, 2) students; see Figure 4). The event and process conceptions were not mutually exclusive and could be activated at the same time. Arguing in the framework of *p*-prims as introduced by diSessa,¹² the former conception is grounded in intuitive physics in the form of a metaphorical breaking of sticks. However, we cannot simply identify the process conception as the more proficient conception. Professional chemists are known to employ a diverse set of

models when talking about the chemical bond.¹ Scientific models are introduced to describe a phenomenon, and in the highly diverse field that chemistry is, it makes sense that these models vary. Hence, we note that all models of chemical bonding carry some valuable information that may prove to be useful depending on the context. For instance, Lewis structures represent the electronic wave function (and can therefore be taken as descriptors for the Born–Oppenheimer potential energy surface).^{53,54}

From a philosophical perspective, Hendry⁵⁵ has come to a similar conclusion when analyzing the conceptions of the chemical bond in a historical context. He identified (i) the structural conception, in which “chemical bonds are, at least for molecular substances, material parts of the molecule that are responsible for spatially localized submolecular relationships between individual atomic centers” (ref 55, p 917) and (ii) the energetic conception, according to which “facts about chemical bonding are just facts about energy changes between molecular or supermolecular states” (ref 55, p 919). While the latter is more agnostic and better compatible with quantum chemistry, it implies explanatory loss, a characteristic highly important in experimental domains.⁵⁵ Likewise, we found the structural conception in a metaphorical language. Bond formation is described as building a bridge, a bond as a chain link and a molecule as an object that can be destroyed (see Table 3). However, the energetic concept was absent in the interview data. This highlights the absence of the concept energy in most (if not all) visual representations. In studies I–III, the attention of the students was specifically guided to the energy, as there was always an energy graph on the screen. In the interview studies, this energy graph was only visible in the last representation with the first three (see Figure 2) being much more typical.

The third theme related the newly learned information about orbitals to prior knowledge of the students. Defining the chemical bond in terms of superposition of valence atomic orbitals is found in the literature.⁵⁶ However, most statements by the students attempting to connect these concepts were inaccurate. For example, the absence of a specific orbital was thought to indicate the absence of a bond, or the presence of the highest occupied molecular orbital at a specific bond was thought to indicate an instability at this location (see Table 2). Again, we did not identify the model as the source of the difficulty, but rather the inability of the students to successfully integrate it into their prior knowledge as a model that is distinct from the others but equally important in the correct context. We found some cases of “Bond is Overlap” in the interview data. However, this metaphor concept only consists of one metaphor and is therefore not very rigorous. Furthermore, this conception might have been prompted by the context in which the students were recruited, the quantum chemistry lecture. The students might have felt obliged to connect the bond concept to the concept of orbitals, which is very prominent in that lecture.

Finally, no students defined bonds in connection with the forces corresponding to the electronic energy gradients that they felt. In this line of reasoning, a bond could have been defined as the absence of force. It was even argued in the literature that a force-based approach might be best to describe the chemical bond in an educational context.^{26,57–59} In studies I–III as well as in the dynamic visual representation in study IV, the bonds were visualized based on interatomic distances and hence appeared not only when the structure was stable.

Moreover, it is not trivial to connect the absence of a stimulus to the presence of a bond. This conception could be supported by not visualizing the bonds at all or only if no resulting force is present.

Finally, we constructed a metaphor concept that was absent in the thematic analysis, “Bond is Community”. Being part of a molecule is belonging, the bond is sharing electrons, bond formation is connecting, and electrons are people (that might be in love). We conjecture that this metaphor concept is missing in the thematic analysis due to the nature of the data. In studies I–III, students mentioned the chemical bond only if they found something contrary to their expectations. In the interviews, the students had a chance to reflect more deeply on the nature of the chemical bond. Furthermore, as we found many occurrences of personification, this metaphor concept might be stronger in spoken language than in written reasoning. In any case, the metaphor concept is strongly supported by the language of expert chemists such as “leaving group”, in German “*Ladung*” for charge which can also mean baggage or “electron pair”. The extent of the metaphor concept in expert language would have to be investigated further to confirm this claim.

Many models are of a metaphorical nature. Del Re called models *idealized experiments* in the sense that they can successfully predict measurable outcomes based on a metaphorical model system.⁶⁰ Many misconceptions that we mentioned in our conceptual framework can be explained by incorrect blending, combining, or overgeneralizing of these metaphorical model systems. For example, the students in Boo’s study applied the metaphorical model system of physical object to the concept of energy requirement, leading them to inaccurately assuming that as constructing a physical object requires energy, so does bond formation.²⁷ Similarly, visualizations of orbitals as physical spaces might potentially create the inaccurate metaphorical mapping of orbitals as a space where electrons are or where electrons move into.⁵

The plurality of conceptions becomes problematic only when the students try to integrate them into a single model, as we have found on multiple occasions. From initial chemistry learning, the students are exposed to this variety of conceptions. With this work, we are not judging the diverse models employed by the students or the instructors but imply that a proper introduction to this diversity without limiting or confusing the students is crucial for successful higher chemistry education.

We have shown that in a single context (that is, initial quantum chemistry learning facilitated by interactive software), multiple models of the chemical bond are still activated, sometimes even simultaneously. Future research should investigate how this affects advanced chemistry learning.

CONCLUSIONS

Implications

Students are taught a number of different models in their higher chemical education. While chemists focus on the models helpful to their specific field, such as hybridization and the octet rule for organic chemists or one-electron wave functions and potential energy hypersurfaces for quantum chemists, the students are exposed to all of them simultaneously. In that process, we suggest that it is crucial to make the students aware of the different models of the same concept and of the different strengths and applications of each

are. One way to support students in their learning is through experiential learning. The real-time interaction with molecules and experience of the forces, which would act on the selected atom, offered in our studies led the students to generally challenge existing conceptions of fundamental quantum chemical concepts, such as the Born–Oppenheimer potential energy hypersurface or minimum-energy structures.³⁸ This experiential quantum chemistry learning warrants further investigation.

Limitations

The data for this analysis were limited in the sense that conceptions of the chemical bond were not the primary focus of a large part of the data collection and, hence, questions focused on the connections between the energy graph and the simulation. In studies I–III, the students were not explicitly asked about their interpretation of the visual representation of the molecules. Accordingly, our data were limited to events in which the students were confused or in disagreement with the visual representation and mentioned this on their questionnaire or in their think-aloud interview. Furthermore, all studies were framed as a testing of the learning environment for future generations of chemistry students. We could therefore expect certain distrust in the representations, including the presence of the chemical bond. While here the chemical bond as a construct that is there or not is highly supported by the visualization, future research should explore how other visualizations, such as the electron density, influence the activated conceptions or how the students would reason about the chemical bond without visual input at all. Finally, expert language seems to have an impact on the metaphorical language employed by students. A systematic analysis of the former would be of interest.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available at <https://pubs.acs.org/doi/10.1021/acs.jchemed.4c00024>.

Study procedure for study IV described in more detail; the analysis processes described in more detail; further data examples including the original German wording (PDF)

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Notes

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